

A12 Docs RAG On Premise

Build a RAG pipeline with Milvus

AI VN Team

5.2026

Agenda

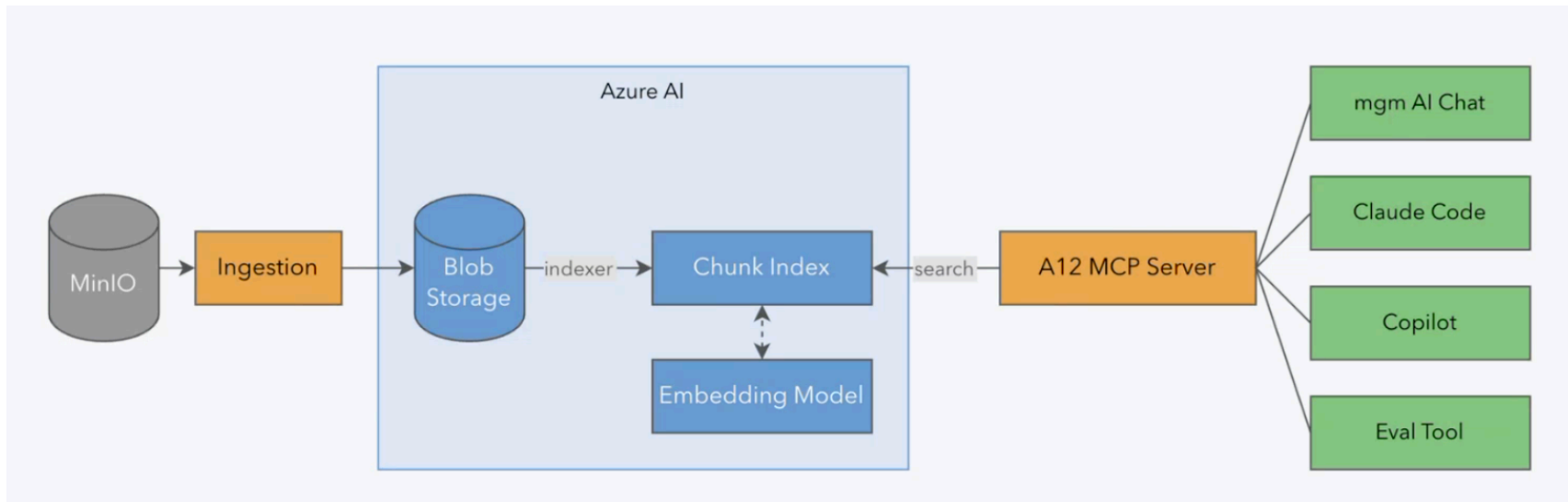
- Goals of moving from Azure cloud to on premise
- Compare architecture solution on Azure cloud and on premise with Milvus
- How to sync data in ingestion on premise
- Compare their chunking strategy
- Compare their retrieval strategy
- Compare their performance
- Leverage relationship of A12 components to help LLM reasoning better
- Suggestions & discussions

Goals of moving from Azure cloud to on premise

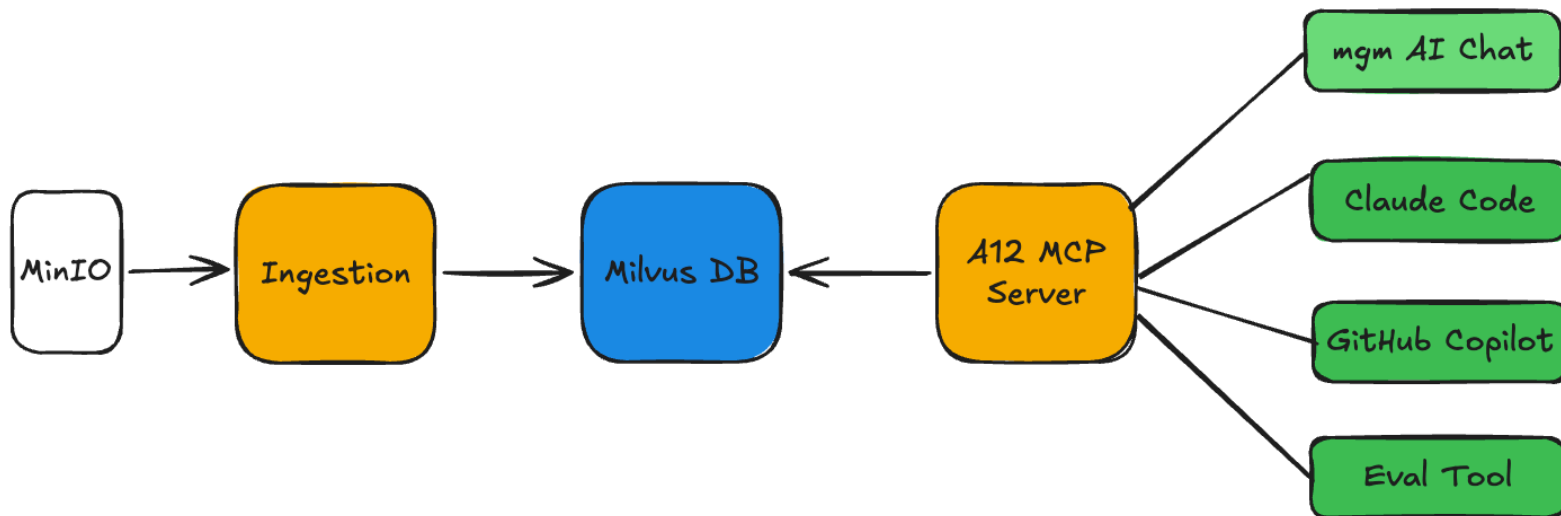
- Minimize dependence on Azure cloud and reduce costs
- Ensure GDPR compliance and data sovereignty on our own infrastructure
- Maintain the same retrieval performance as A12 RAG on Azure
- Make it easy to control and address nuanced problems in A12 RAG

Overview A12 RAG MCP architecture on Azure cloud

- Azure AI search

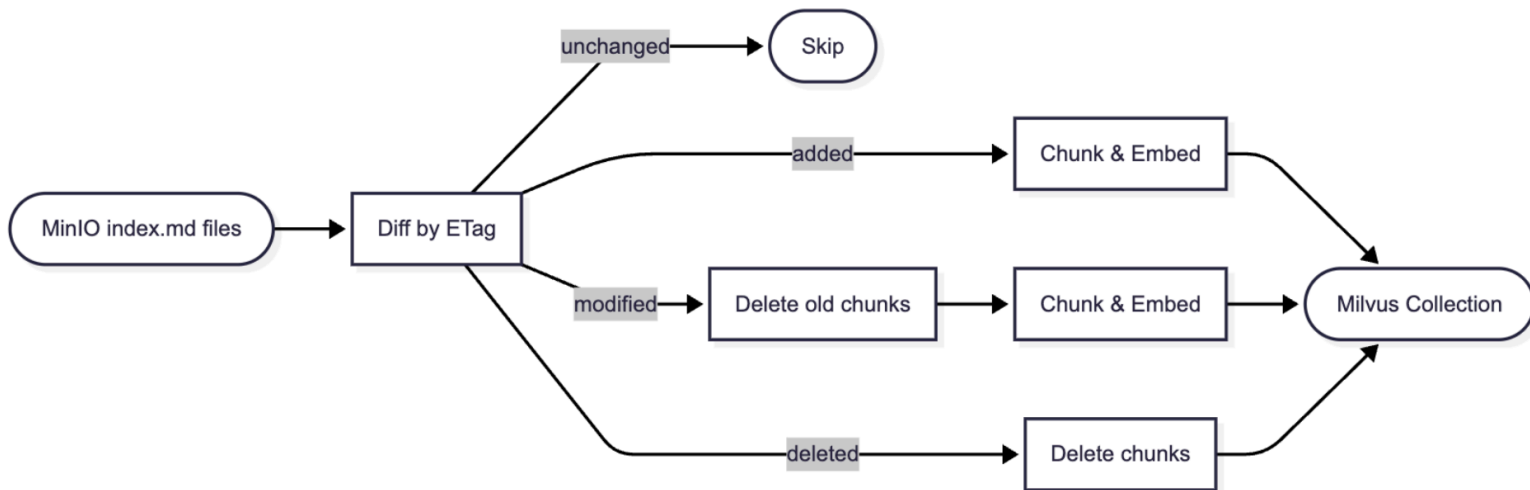


Overview A12 RAG MCP architecture on premise



- Replace Azure AI Search with Milvus
- Remove Azure Storage dependency
- Keep Azure OpenAI for text-embedding-3-large

A12 Docs ingestion on premise (1)



- Ingestion still runs on a schedule
- Milvus manages data sync from MinIO
- Each A12 doc version maps to a partition in the Milvus collection — ANN search is scoped to one or more partitions explicitly

A12 Docs ingestion on premise (2)

- Sample logs during data sync

```
Partitions: 2023_06_ext11
Total chunks: 3729
To sync: 2025.06/ext4/*
```

```
To add 112
To modify 0
To delete 0
Unchanged 0
```

```
Version 2025.06-ext4 (0 deleted 0 modified 112 added)
```

```
[2/112] 2025.06/ext4/utils_logging_collections/utils-logging-documentation/index.md → 18 chunks
[3/112] 2025.06/ext4/cms/cms-dev-docs/index.md → 21 chunks
[4/112] 2025.06/ext4/utils_logging_collections/utils-collections-documentation/index.md → 7 chunks
[1/112] 2025.06/ext4/base/base-documentation-bundle/index.md → 16 chunks
```

A12 Docs ingestion on premise (3)

default (3)

a12_docs (3,729)

a12_sync_jobs (1)

a12_sync_state (58)

Collections Properties

+ Collection

+ Import File

Refresh

Search by name

☐	Name	Status	Approx Entity Count ^②	Description	Alias ^②	Created At	Updated At
☐	a12_docs	Loaded	3,729	--	-- +	5/4/2026, 4:29:19 PM	5/4/2026, 4:29:19 PM
☐	a12_sync_jobs	Loaded	1	--	-- +	5/4/2026, 4:29:21 PM	5/4/2026, 4:29:21 PM
☐	a12_sync_state	Loaded	58	--	-- +	5/4/2026, 4:29:20 PM	5/4/2026, 4:29:20 PM

- a12_docs : stores document chunks
- a12_sync_jobs : tracks ingestion jobs
- a12_sync_state : tracks synced document files

Chunking Challenges: New features in roadmap

- Assets [PRIO 1]
 - Statistic: What assets exist?
 - Add general pre-process functionality between download from MinIO and Upload to Azure Blob.
 - First preprocess: Move link processing from retrieval to ingestion
 - Inline tables [PRIO 0]
 - Inline short code snippets
- Observability: Trinh <https://jira.mgm-tp.com/jira/browse/DS-2027?jql=component%20%3D%20a12-rag>
- Publish Docker Image to Artifactory [PRIO 1]

Find Concept

- **Prepare for external usage [PRIO 0, June Release]**
 - Prepare for public vs enterprise source bucket (see <https://jira.mgm-tp.com/jira/browse/A12DOC-1057>)
 - Auth (GetA12 Identity Token?)  [DS-2064](#) - a12-rag-mcp: Authenticate A12 RAG MCP Server **IN PROGRESS**
 - Security Review (SP, Benjamin Weller)
- Code Example MCP Tool [PRIO 2]
 - Find: high-quality code snippets. Source Ideas:
 - Showcases
 - A12 Source Code + Test Codee
 - A12 Gems
- Model Example MCP Tool [PRIO 2]
- Assets (Images, HTML, Code) [PRIO 3]
 - Large Code snippets? (Code more difficult than prose)
 - Transcribe Images (Azure) and inline (with link to img)
- Move MCP Deployment to C12 Cluster [PRIO 2]

Chunking Challenges in A12 Docs (1)

- **Problem 1:** A12 Docs frequently contain intra-references — links pointing to other sections within the same document

2023.06-ext11
🔍

- Get Started
- Tutorials
- Components**
 - Base
 - Client
 - Data Distribution
 - Data Services**
 - Engines
 - Expressions
 - Kernel
 - Model Editor (SME)
 - Notification Center
 - UAA

Data Services

The most common use-cases follow:

Single Server

If you run just a single Data Services server, you can use embedded Active MQ broker to manage the queue. To achieve this, use the Asynchronous CDD Spring profile together with the Embedded JMS Spring profile.

This is primarily intended for development and testing environments, and it is also possible to run CDD processing synchronously. Please see the Synchronous CDD Spring profile and pay attention to the Synchronous CDD Processing section below.



Keep in mind, that synchronous processing could be a performance bottleneck in a production environment.

Chunking Challenges in A12 Docs (2)

- The RAG pipeline cannot answer questions that depend on content from these intra-references

Question Details

What is required to enable CDD processing on a single server?

dataservices

easy

Evaluation Metrics

Answer Relevance ⓘ

0.643

Answer Correctness ⓘ

0.548

Context Precision ⓘ

0.589

Context Recall ⓘ

0.000

Expected Answer

For a single server setup, an embedded Active MQ broker can be used with the following configurations: `spring.artemis.mode=embedded`, `spring.artemis.embedded.enabled=true`, and enabling the producer and consumer with appropriate properties.

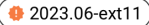
Generated Answer

To enable CDD processing on a single server, you need to:

1. Use the **Asynchronous CDD** Spring profile (`spring-profile-cdd\_async`) along with the **Embedded JMS** Spring profile (`spring-profile-embedded\_jms`) to manage the dirty-documents queue with an embedded ActiveMQ broker.
2. Set the following configuration properties:
 - `mgmtp.a12.dataservices.cdd.dirtyDocuments.async.consumer.enabled=true`
 - `mgmtp.a12.dataservices.cdd.dirtyDocuments.async.producer.enabled=true`

Chunking Challenges in A12 Docs (3)

- **Problem 2:** A12 Docs contain long code blocks that exceed chunk size limits




Helm A12 Stack

This is `values.yaml` example. You must change `127-0-0-1` to the IP of your cluster, and `storageClass` to the storage class of your cluster.

```

1 a12-secrets:
2   dockerConfigJson:
3     enabled: true
4     username: admin
5
6   postgresql:
7     enabled: true
8   dataServices:
9     enabled: true
10  camunda:
11    enabled: true
12  contentStore:
13    enabled: true
14
15  hazelcastManagementCenter:
16    enabled: true
17    username: admin
18
19  jwt:
20    enabled: true
21
22  workflows:
23    restClient:
24      enabled: true
25      username: admin

```

Chunking Challenges in A12 Docs (4)

- **Problem 3:** A12 Docs contain long table blocks that span multiple chunks

2023.06-ext11

Workflows

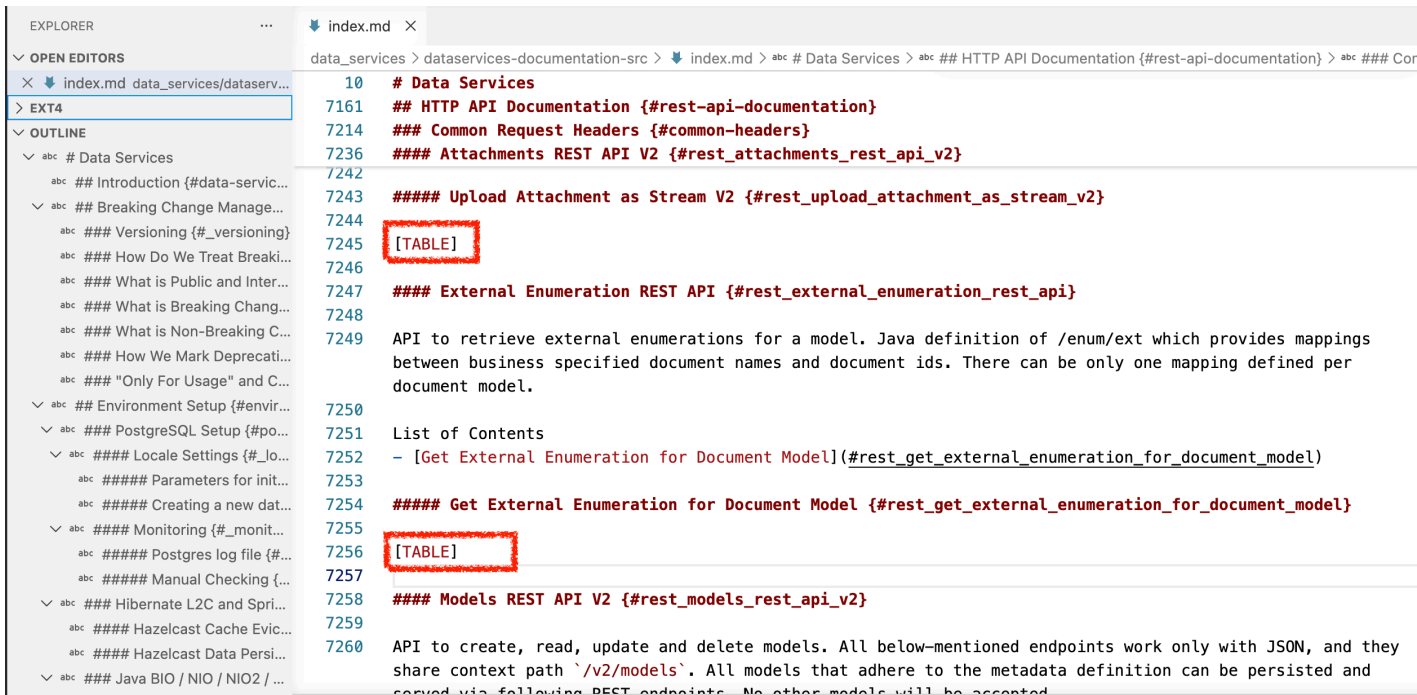
Overall
Modeling
Development
Get Started
Tutorials
Components
Base
Client
Data Distribution
Data Services
Engines
Expressions
Kernel
Model Editor (SME)
Notification Center
UAA
Utils
Widgets
Workflows

Scopes for Process-Related Authorization Rules

Endpoint	Arguments available in authorization file	Scope
Endpoint to start the process: <code>/api/process-definition/{key}/start</code>	<code>#processDefinitionKey:</code> String <code>#inputDocuments:</code> List<InputDocumentDT0>	ProcessDefinitionCreateInst: - runs before entering the method
Endpoint to get list of process definitions: <code>/api/process-definition</code>	-	ProcessDefinitionRead - runs before entering the method
	<code>#resource:</code> ProcessDefinition	ProcessDefinitionFilter - runs after method execution and allows filter list of process definitions, by applying defined rules to each elen in the list
Endpoint to get process diagram: <code>/api/process-definition/{key}/diagram</code>	<code>#processDefinitionKey:</code> String	ProcessDefinitionRead - runs before entering the method

Chunking Challenges in A12 Docs (5)

- **Problem 4:** A12 Docs contain table placeholders in Markdown that are replaced during AsciiDoctor builds



The screenshot shows an IDE with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with folders like 'data_services' and 'dataservices-documentation-src'. The code editor displays a Markdown file with the following content:

```

10 # Data Services
7161 ## HTTP API Documentation {#rest-api-documentation}
7214 ### Common Request Headers {#common-headers}
7236 #### Attachments REST API V2 {#rest_attachments_rest_api_v2}
7242
7243 ##### Upload Attachment as Stream V2 {#rest_upload_attachment_as_stream_v2}
7244
7245 [TABLE]
7246
7247 ##### External Enumeration REST API {#rest_external_enumeration_rest_api}
7248
7249 API to retrieve external enumerations for a model. Java definition of /enum/ext which provides mappings
between business specified document names and document ids. There can be only one mapping defined per
document model.
7250
7251 List of Contents
7252 - [Get External Enumeration for Document Model] {#rest_get_external_enumeration_for_document_model}
7253
7254 ##### Get External Enumeration for Document Model {#rest_get_external_enumeration_for_document_model}
7255
7256 [TABLE]
7257
7258 ##### Models REST API V2 {#rest_models_rest_api_v2}
7259
7260 API to create, read, update and delete models. All below-mentioned endpoints work only with JSON, and they
share context path `/v2/models`. All models that adhere to the metadata definition can be persisted and
saved via following REST endpoints. No other models will be accepted

```

Two instances of the placeholder `[TABLE]` are highlighted with red dashed boxes, indicating where table content is missing or to be replaced during the build process.

Compare their chunking strategy

AZURE AI SEARCH	MILVUS SEARCH	STEP TYPE	PURPOSE
Split by headings (h1-h3)	Split by headings (h1-h4)	Common	-
Chunk by character limit (2000 chars, 500 overlap)	Chunk by recursive splitter, cap = 3000 chars, no overlap	Common	-
None	Append breadcrumb to chunk text	Specific	Separates similar text across different components; anchors chunks to their source document context
None	Detect inline code and tables	Specific	Future: preserve table headers across chunks, help the LLM retrieve full blocks via tools
None	Detect intra-references in chunks	Specific	Future: return referenced content within the chunk
None	Detect sections spanning multiple chunks	Specific	Provide a tool to retrieve the full section

Compare their chunking strategy - Chunk text in Milvus

default (3) / a12_docs

default (3)

- a12_docs (14,858)
- a12_sync_jobs (5)
- a12_sync_state (267)

Schema Data Vector Search Partitions Segments

Query Expression

Please enter your query expression, eg: id > 0

1 selected Import File Insert Sample Data Empty

is_overflow	section_id
UE	5d01efa4-ee9e-461a-b9a2-08d1c7e693fa
LSE	a7c8cb46-1fcd-4636-99ff-21e14889232a
LSE	9055fd8f-af10-437e-8031-37a842dbebeb
LSE	6ee9eb95-7ca1-42fb-a8a9-272d184e1005
UE	d5798bdb-9ed0-413c-9d6e-4e2324c4361e
UE	e95e0a2e-6266-4ae2-85b2-183593d07d0c
UE	1662836e-9172-4c8c-b535-15459f513baa
UE	023ed750-ef6d-42ec-a7ee-15f65727b235
LSE	5a07f2cb-b0b8-4b45-98c3-1a3bbd14ef5c
LSE	51aaf994-e1b4-4438-b11f-5a89e03bc597

Entity Data

parent_id

61a58d23-1ee1-4d5f-8a8c-3ad2209ba37a

text

A12 Data Services > [](#json-rpc-endpoint){.anchor}JSON-RPC Endpoint > JSON-RPC 2.0 Core Operations > ADD_LINK

ADD_LINK

- This operation does not ensure the deferred data integrity constraints (upper limit). Instead, the ADD LINK operation fires events that fill a ThreadLocal collection, which is used by 'RelationshipLinkOperationValidator' only after all the operations are finished. Calling this operation directly therefore might corrupt link integrity and cause issues with subsequent RPC operations due to the usage of ThreadLocal collections for validation.

Call sequence {#_call_sequence_14}

![link addLinkOperationSequence]
(https://geta12.com/docs/2023.06/ext11/data_services/dataservices-documentation-src/assets/images/link_addLinkOperationSequence.svg)

Figure 14. Sequence of ADD_LINK calls.

```

| | |
|---|---|
| | This sequence is compliant with the default settings without caches and with rpc
enabled which should show most of the events and security checks. Using different

```


Enhance chunking strategy - Consider aspects

ASPECT	AZURE AI SEARCH	ON PREMISE
Extend chunking logic	Difficult — requires additional Azure services (e.g. Azure Functions)	Easy — just change the code
Cost	Higher cost	Free
Dependencies	More dependencies	No additional dependencies

Compare retrieval strategy

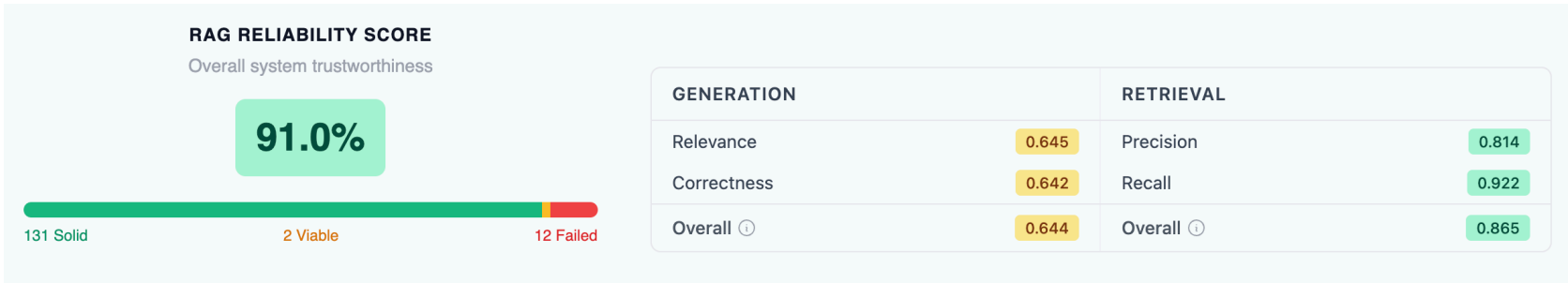
AZURE AI SEARCH	MILVUS SEARCH
Hybrid search (ANN search = 50, BM25 = 1000)	Hybrid search (ANN search = 50)
RRF (topK = 50)	RRF
Built-in semantic reranker	None

Reranker flexibility

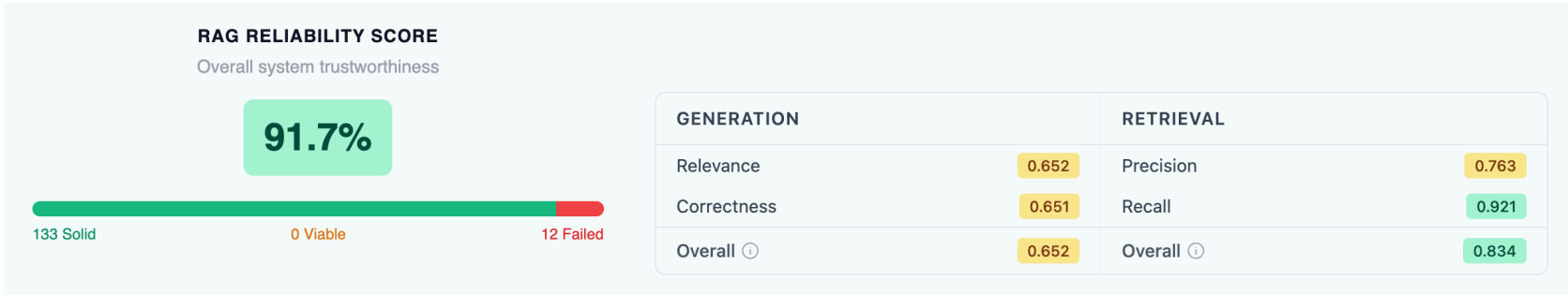
- **Azure AI Search:** the semantic reranker is built-in and optional — no alternative rerankers are available
- **On-premise:** choose from Milvus's built-in rerankers or implement a fully custom reranker

Compare A12 RAG Evaluation

- Azure AI Search



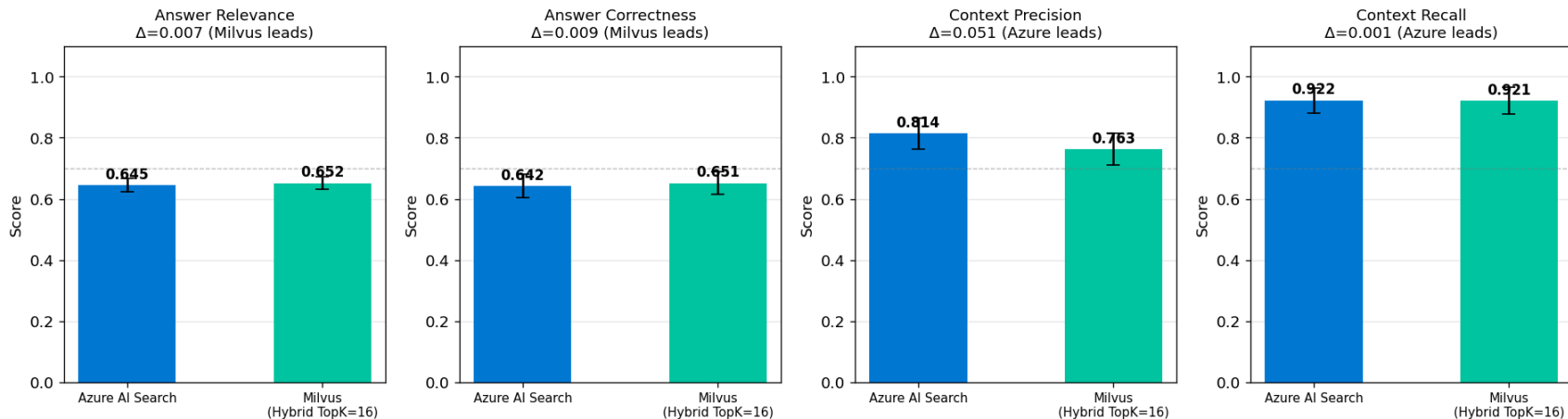
- Milvus search



Evaluation Insights between Azure AI Search and Milvus search



Mean Score per Metric — Azure AI Search vs Milvus Hybrid (TopK=16)
error bars = 95% CI



Failed questions between two searches – Statistics

Failure = `context_precision < 0.5` **AND** `context_recall < 0.5`

CATEGORY	BOTH	AZURE_ONLY	MILVUS_ONLY	TOTAL_FAILED	TOTAL_QUESTIONS	FAILURE_RATE
dataservices	0	0	0	0	25	0.0%
dev	0	0	2	2	30	6.7%
plasma	0	1	0	1	30	3.3%
quickstart_guide	4	0	1	5	30	16.7%
uaa	1	2	0	3	30	10.0%
TOTAL	5	3	3	11	145	7.6%

- `quickstart_guide` is the hardest category for both engines (16.7% failure). Most of their failures due to [TABLE] placeholder and bad groundtruth
- The 3 azure-only and 3 milvus-only failures are complementary – each engine recovers what the other misses

Failed questions between two searches (1)

CATEGORY	FAILED_IN	QUESTION_TEXT	GROUND_TRUTH	ROOT CAUSE
quickstart_guide	both	What are the user roles that are available in all workspaces?	The roles admin, systemAdmin, guest and userManagementAdmin are available in all workspaces.	[TABLE] placeholder
quickstart_guide	both	What are the user roles that are not available in all workspaces?	The roles customer, shop and showcase are only available in the workspace e-commerce.	[TABLE] placeholder
quickstart_guide	both	What is the Data Modeler?	An older tool used to create and edit Document Models. This tool is deprecated and retired in new A12 versions.	Many relevance chunks
quickstart_guide	both	Can you give me an example of heterogeneity?	An example of heterogeneity can be found in the e-commerce workspace in the products module where a heterogeneous list of products is shown.	Bad groundtruth
uua	both	How can I disable CORS during authentication?	You can set the property <code>mgmtp.a12.uua.authentication.cors.enable</code> to false.	[TABLE] placeholder

Failed questions between two searches (2)

CATEGORY	FAILED_IN	QUESTION_TEXT	GROUND_TRUTH	AZURE_CR	MILVUS_CR
uaa	azure_only	Which Localizer does the UAA javascript-client use?	The UAA client uses a Localizer from <code>@com.mgmp.a12.utils/utils-localization</code> as localizer props or by useContext	0.0	1.0
uaa	azure_only	What API function from UAA can I use to keep information between tabs?	UAA uses the UaaSecureStorage to store information, so if you want transfer information between tabs you can use the function transferSessionStorage.	0.0	1.0
plasma	azure_only	What are the three Plasma Design Principles?	The three Design Principles are Robustness, Efficiency and Accessibility	0.0	1.0

Failed questions between two searches (3)

CATEGORY	FAILED_IN	QUESTION_TEXT	GROUND_TRUTH	AZURE_CR	MILVUS_CR
dev	milvus_only	How can validation be disabled for specific document models?	To disable validation for specific document models, use the <code>mgmtp.a12.workflows.skipPartialValidationForModels</code> property for task updates and completions. To fully disable validation, use the <code>mgmtp.a12.dataservices.documents.validation.skipForModels</code> property in Camunda Service.	0.5	0.0
quickstart_guide	milvus_only	What does BAP stand for?	Business Application Platform.	1.0	0.0
dev	milvus_only	What attachment handling is supported by A12WF?	A12WF only supports inline attachments using the provided <code>inlineAttachmentHandler</code> .	1.0	0.0

Leveraging A12 component relationships to improve search

- A12 docs are technical documents with strong dependencies between components.

LAYER	NAME	COMPONENTS
0	Utilities & Foundation	utils_localization, utils_logging_collections, utils_server_connector, base, plasma
1	Core Modeling Foundation	kernel, expression
2	Design-Time (Modeling)	sme, diagram_editor, tdg
3	Server Runtime	data_services, uaa, user_management, transformer, data_distribution, workflows
4	Client Runtime	client, form_engine, overview_engine, tree_engine, crud, relationship_engine, content_engine, notification_center
5	UI & Output	widgets, print_engine, cms
—	Cross-cutting	build_and_deployment, project_template, overall

Leveraging A12 component relationships — MCP instruction

BEFORE

Use `search_a12_docs` to search the A12 knowledge base (geta12.com). A12 is a model-driven enterprise application platform. The knowledge base contains versioned official documentation covering modeling, client/server runtime, design system, deployment, and more.

AFTER

Use `search_a12_docs` to search the A12 knowledge base (geta12.com). A12 is a model-driven enterprise application platform with ~28 components organized in 6 architectural layers plus cross-cutting components:

- **Layer 0 — Utilities & Foundation:** `utils_localization`, `utils_logging_collections`, `utils_server_connector`, `base`, `plasma`
- **Layer 1 — Core Modeling Foundation:** `kernel`, `expression`
- **Layer 2 — Design-Time (Modeling):** `sme`, `diagram_editor`, `tdg`
- **Layer 3 — Server Runtime:** `data_services`, `uaa`, `user_management`, `transformer`, `data_distribution`, `workflows`
- **Layer 4 — Client Runtime:** `client`, `form_engine`, `overview_engine`, `tree_engine`, `crud`, `relationship_engine`, `content_engine`, `notification_center`
- **Layer 5 — UI & Output:** `widgets`, `print_engine`, `cms`
- **Cross-cutting — Lifecycle & Onboarding:** `build_and_deployment`, `project_template`, `overall`

Leveraging A12 component relationships — Tool description

BEFORE

Search A12 platform documentation via semantic search. Returns ranked document chunks with scores. Supports filtering by A12 version. Does not generate answers. Returns raw documentation excerpts only.

AFTER

Search A12 platform documentation via semantic search. Returns ranked document chunks with scores. Supports filtering by A12 version **and component**. Does not generate answers. Returns raw documentation excerpts only.

Component Search Strategy — If the first search surfaces an unexpected component, use the cluster list below to pick 1–2 related components for follow-up searches via the `a12_component` parameter. Clusters are intent-based, not exclusive — a component may belong to multiple clusters.

Key clusters — Frontend (UI rendering): client, form_engine, overview_engine, widgets · Modeling pipeline (how-to-model): sme, kernel, form_engine · Backend/API (server logic): data_services, uaa, workflows · Workflow (BPMN tasks): workflows, form_engine, data_services, client · Notifications (alerts/messaging): data_distribution, notification_center · Deployment (infra/setup): build_and_deployment, project_template, uaa, data_services · Getting started (orientation): overall, project_template

Claude Code with A12 RAG MCP in Azure

Sample prompt: In A12 Platform, how do I create a model like Student and manage them in table.

- Tool call 1

```
> In A12 Platform, how do I create a model like Student and manage them in table.
```

- Calling a12-docs-azure-ai-search 2 times... (ctrl+o to expand)

use-a12-rag-mcp-with-azure-ai-search

Tool use

```
a12-docs-azure-ai-search - search_a12_docs(question: "create document model entity with fields step by step",
top_k: 5) (MCP)
```

Search A12 platform documentation via semantic search. Returns ranked document chunks with scores. Supports filtering by A12 version. Does not generate answers. Returns raw documentation excerpts only.

Do you want to proceed?

```
> 1. Yes
```

```
2. Yes, and don't ask again for a12-docs-azure-ai-search - search_a12_docs commands in
/Users/ttdinh/Documents/Working/Presentation/A12_RAG/present-milvus-search
```

```
3. No
```

- Tool call 2

```
> In A12 Platform, how do I create a model like Student and manage them in table.
```

- Calling a12-docs-azure-ai-search 3 times... (ctrl+o to expand)

use-a12-rag-mcp-with-azure-ai-search

Innovation Implemented.

Claude Code with A12 RAG MCP in Azure (2)

- Answer

- In A12, "creating a Student and managing them in a table" is done by combining **three models** in the Simple Model Editor (SME). There's no code – it's all modeling.

1. Document Model – **Student_DM** (the data)

This defines the entity's structure.

- Workspace Explorer → **Add** → **Document Model** → name **Student_DM**, choose locales/roles.
- Add a root **Group** named **Student**.
- Inside it, **Add** → **Field** for each attribute, e.g.:
 - **FirstName** (String)
 - **LastName** (String)
 - **DateOfBirth** (Date)
 - **MatriculationNumber** (String)
 - **Status** (Enumeration: ACTIVE / GRADUATED / ...)
- Save.

2. Form Model – **Student_FM** (create / edit a single Student)

- **Add** → **Form Model**, bind it to **Student_DM**.
- Add a Control Grid + Rows, then drag fields from the Data Models panel into rows.
- Add **Save** and **Cancel** buttons at the bottom.

3. Overview Model – **Student_OM** (the table)

This is the table view that lists and manages all students.

- **Add** → **Overview Model**, select **Student_DM** as the Document Model.
- **Overview tab** → **Columns** → **Add** a Reference column per field you want shown (e.g. FirstName, LastName, MatriculationNumber, Status). Mark each **Sortable** if desired. Reorder via drag-and-drop.
- **Custom Actions tab**:

Claude Code with A12 RAG MCP on premise

- Tool call 1

```
> In A12 Platform, how do I create a model like Student and manage them in table.
```

Calling a12-docs-milvus-search 2 times... (ctrl+o to expand)

use-a12-rag-mcp-with-milvus-v3

Tool use

```
a12-docs-milvus-search - search_a12_docs(question: "How to create a new entity model like Student and display/manage records in a table", top_k: 8) (MCP)
```

Search A12 platform documentation via semantic search. Returns ranked document chunks with scores. Supports filtering by A12 version and component. Does not generate answers. Returns raw documentation excerpts only.

```
## Component Search Strategy...
```

Do you want to proceed?

```
> 1. Yes
```

```
2. Yes, and don't ask again for a12-docs-milvus-search - search_a12_docs commands in /Users/ttdinh/Documents/Working/Presentation/A12_RAG/present-milvus-search
```

```
3. No
```

- Tool call 2

```
> In A12 Platform, how do I create a model like Student and manage them in table.
```

Calling a12-docs-milvus-search 2 times... (ctrl+o to expand)

use-a12-rag-mcp-with-milvus-v3

Tool use

Claude Code with A12 RAG MCP on premise (2)

- Answer

› In A12 Platform, how do I create a model like Student and manage them in table.

Called a12-docs-milvus-search 2 times (ctrl+o to expand)

- In A12, "managing entities like Student in a table" is the canonical use case for the platform. You build it by composing **four models** in the **Simple Model Editor (SME)** – no code required. Here's the workflow:

1. Document Model – Student_DM

The blueprint of what data you persist. In SME: **Add → Document Model**, name it **Student_DM**, then add fields (e.g. `firstName: String, lastName: String, dateOfBirth: Date, studentId: String, email: String`).

2. Form Model – Student_FM

Defines how the Document Model is collected/edited as a form. **Add → Form Model**, set Document Model to **Student_DM**. Drag fields onto the form, group them, add **Save** and **Cancel** buttons in the footer. This is what users see when creating or editing a single Student.

3. Overview Model – Student_OM

This is the **table view**. **Add → Overview Model**, set Document Model to **Student_DM**. Configure:

- **Columns** – pick which **Student_DM** fields appear (e.g. `studentId`, `name`, `email`)
- **Features** – pagination, search, sort, filter, *Show Number of Entries*
- **Row actions** – e.g. Edit / Delete buttons
- **Toolbar actions** – e.g. *Create Student* button

4. Application / Module Model – Student_AM

Wires **Student_FM** and **Student_OM** together. The simplest pattern is the **Master-Detail Module Model**: the overview (table) is the master, the form is the detail panel that opens when a row is selected or *Create* is clicked.

Suggestions & discussions

- Suggestions
 - Convert the code from TypeScript to Python
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